Drone Swarms for Post-Disaster Search and Rescue in Remote and Inaccessible Areas

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***Abstract*—Drone swarms offer a transformative approach to post-disaster search and rescue (SAR), particularly in remote and inaccessible areas where traditional methods are limited by terrain, infrastructure collapse, or communication failure. This paper investigates the deployment of autonomous drone swarms equipped with thermal imaging, LiDAR, and computer vision sensors for real-time mapping and survivor detection. A decentralized control architecture, inspired by biological swarming and supported by multi-agent reinforcement learning, enables coordination in GPS-denied and dynamic environments. Edge computing and mesh networking ensure low-latency communication and operational resilience. Field experiments demonstrate significant improvements in area coverage, detection accuracy, and response time when compared to conventional SAR or single-drone systems. The integration of these technologies provides a scalable and adaptive SAR platform that minimizes risks to human rescuers. This study advances the development of intelligent aerial systems for humanitarian applications, enabling faster and safer responses in the aftermath of natural and man-made disasters.**

Keywords—Swarm, Machine learning, Artificial intelligence, Communication strategies

I. Introduction

Natural and man-made disasters—including earthquakes, tsunamis, landslides, hurricanes, industrial explosions, and armed conflicts—present extreme challenges for traditional emergency response teams, particularly in geographically isolated or infrastructurally damaged regions. The "golden hours" immediately following a disaster are critical for maximizing survival rates; however, ground teams often face accessibility barriers due to destroyed roads, unstable structures, and environmental hazards [1]. Unmanned Aerial Vehicles (UAVs), specifically drone swarms, offer an intelligent, autonomous, and scalable solution to augment search and rescue (SAR) operations in such scenarios.

Drone swarms—comprising multiple cooperative UAVs working as a decentralized, coordinated unit—can provide persistent aerial surveillance, terrain mapping, survivor detection, and supply delivery. Unlike single UAVs, swarms operate collaboratively through swarm intelligence principles, enabling them to search vast or occluded environments efficiently, adapt to real-time conditions, and continue functioning despite individual drone failures [2]. This approach leverages bio-inspired algorithms such as particle swarm optimization (PSO), ant colony optimization (ACO), and flocking behaviors to replicate the emergent intelligence observed in social insects or birds [3].

A. Technological Enablers of Drone Swarms

Recent advancements in edge computing, wireless mesh networking, multi-agent systems, and artificial intelligence (AI) have driven the viability of drone swarms for SAR applications. Each UAV in the swarm is typically equipped with high-resolution cameras, thermal sensors, LiDAR, and onboard processing units that allow for real-time data analysis and decision-making at the edge [4].

Communication among drones and between the swarm and the ground station is established via multi-hop ad hoc networks (e.g., FANETs – Flying Ad Hoc Networks), ensuring resilience against node failure or signal obstruction in disaster zones [5]. Reinforcement learning (RL)-based swarm controllers further enhance adaptability in unstructured environments, allowing drones to optimize their path planning, coverage, and target detection strategies without central supervision [6]. Figure 1 shows layered communication (UAV–UAV, UAV–Ground, UAV–Cloud), onboard sensors (thermal, LiDAR, visual), and decentralized coordination modules.

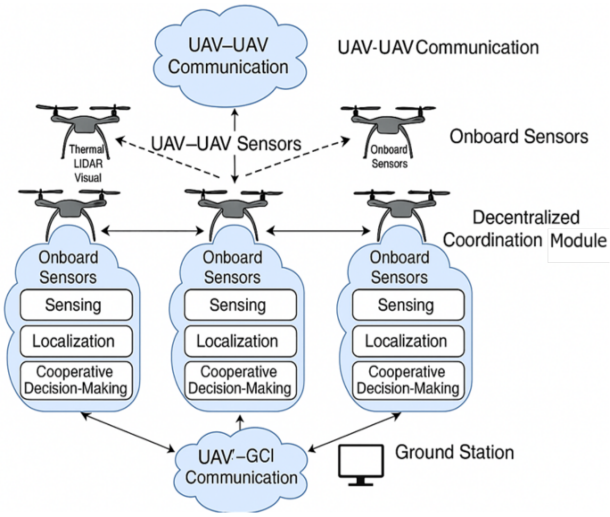

Fig. 1. Swarm Architecture for Post-Disaster SAR Missions

B. Importance in Remote and Inaccessible Areas

Remote and topographically challenging regions—such as mountainous zones, isolated islands, collapsed urban infrastructures, or dense forests—amplify the logistical constraints of traditional SAR operations. In these environments, drone swarms can traverse impassable terrain, fly below radar for site mapping, and relay high-fidelity spatial and thermal data to incident command centers [7]. Additionally, drones can deploy micro-medical kits, water, or communication devices to stranded victims, bridging critical response gaps until ground crews arrive.

In recent disasters like the 2023 Turkey-Syria earthquakes and the 2020 Beirut explosion, UAVs were deployed in isolated zones to assess structural damage and detect

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survivors. However, these missions relied on limited, single-drone operations lacking autonomy and scalability. The transition to swarm-based models represents a paradigm shift—moving from reactive, manually piloted systems to proactive, intelligent, and distributed aerial responders [8]. Table 1 shows the representative drone swarm systems for SAR operations.

Table 1. Review of State-of-the-Art Drone Swarm Systems in SAR

System/ Project	Country	Technology Stack	SAR Role	Key Features
DARPA OFFSET [9]	USA	Multi-UAV autonomy, 3D mapping, RL swarm logic	Urban SAR, exploration	Scalable swarm architecture, virtual testing
SMAVNET [10]	Switzerland	Fixed-wing micro drones, GPS-decentralized comms	Indoor survivor detection	Self-organizing swarm in GPS-denied areas
FLY4SAR [11]	EU (France-led)	AI-enhanced flight, collaborative mapping	Rapid SAR in rural terrain	AI coordination, real-time coverage planning
TDRS by NDRCC [12]	China	Infrared, LiDAR drones with formation flight	Flood rescue, debris scanning	Night operation, water navigation capability

C. Research Gaps and Need for Advanced Swarm Systems

Despite notable progress, existing swarm implementations for SAR lack real-time coordination in dynamic obstacle-rich environments. Issues like communication latency, suboptimal energy management, data overload, and incomplete area coverage limit their deployment scalability [13]. Moreover, ethical and regulatory frameworks remain ambiguous for autonomous swarm operations in civilian airspace during crises [14].

This research aims to advance a robust, adaptive, and scalable drone swarm architecture capable of autonomous deployment in post-disaster scenarios—particularly those marked by terrain inaccessibility and critical urgency. It proposes new algorithms, network protocols, and hardware configurations to improve real-time victim detection, swarm survivability, and operational effectiveness in the field.

II. Literature Review

Unmanned Aerial Vehicles (UAVs) have undergone a significant evolution since their inception in military reconnaissance, progressing into applications across commercial delivery, agriculture, and increasingly, humanitarian response. In the aftermath of disasters such as the 2010 Haiti earthquake and the 2011 Tōhoku tsunami, UAVs began to emerge as vital tools for rapid damage assessment, communications relay, and search-and-rescue (SAR) operations [14]. The limitations of single-drone systems, including limited battery life, restricted field of view, and communication failure under GPS-denied conditions, have led to growing interest in drone swarms that emulate collective intelligence and distributed control found in nature [1].

Swarm robotics, inspired by biological collectives such as ant colonies and bird flocks, emerged in the late 1990s with an emphasis on decentralized coordination and robustness to individual unit failure. Early efforts focused on theoretical models using cellular automata and simple behavior rules [15]. The work of Beni and Wang (1989) [16] introduced the concept of swarm intelligence for robotic systems, laying the foundation for the development of real-world swarm UAV platforms capable of executing complex cooperative tasks. The introduction of behavior-based control, leader-follower models, and consensus algorithms during the 2000s enabled rudimentary swarm deployments in indoor and controlled outdoor environments [17].

A. Swarm Algorithms and Control Architectures

Several categories of algorithms have been instrumental in swarm control:

- Flocking algorithms: Inspired by Reynolds' model, these ensure collision avoidance, alignment, and cohesion [18].
- Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO): Useful in area coverage and path optimization under uncertain terrain conditions [19].
- Distributed consensus algorithms: Allow drones to agree on shared information such as the location of victims or area maps [20].

Hybrid strategies combining PSO with Deep Reinforcement Learning (DRL) are also emerging, allowing adaptive mission planning in dynamic post-disaster conditions [21].

Table 2. Key Milestones in Drone Swarm Development for SAR

Year	Project/ System	Description	Country
2009	SMAVNET	Fixed-wing micro-UAVs using ad hoc communication in collapsed structures	Switzerland
2015	DARPA OFFSET	Tactical drone swarms for urban combat simulations	USA
2018	SkyBorg Project	Autonomous aerial platform for adaptive missions	USA
2020	FLY4SAR	AI-enhanced UAV swarm for rural disaster recovery	EU
2021	TDRS (NDRCC)	Drone swarm for flood SAR with LiDAR/IR sensing	China

B. Sensor Technologies Enabling Swarm Autonomy

Modern drone swarms for SAR are equipped with advanced sensing capabilities that support autonomous exploration and victim detection:

- Thermal imaging: Crucial for detecting heat signatures of survivors under debris [10].
- LiDAR: Enables 3D mapping of collapsed infrastructure [21].
- Multispectral imaging: Helps identify vegetation cover or water bodies [22].
- Visual cameras with CNNs: Allow real-time object detection, person recognition, and terrain segmentation [23].

C. Communication Protocols in Swarm Networks

Drone swarms rely on multi-hop ad hoc networks, including Flying Ad Hoc Networks (FANETs), which must be robust to node mobility and environmental interference. Mesh topology and Software-Defined Networking (SDN) have been explored to maintain Quality of Service (QoS) and low latency [24]. Communication failure mitigation using redundant routing and opportunistic transmission has also been proposed for environments like collapsed urban areas [25].

Energy remains a bottleneck for prolonged swarm operations. Algorithms for energy-aware path planning, wireless recharging stations, and energy-harvesting UAV wings are currently under study [26]. Techniques like scheduling drones for active/passive states and deploying heterogeneous UAVs with relay drones have proven effective in extending mission time [27].

The application of machine learning, especially reinforcement learning and federated learning, has enabled swarm UAVs to make decentralized decisions without relying on constant uplinks. Models trained using simulators like AirSim or Gazebo help transfer policies to real-world deployment [28].

Deploying autonomous swarms in public airspaces raises ethical and legal concerns related to privacy, airspace regulation, and accountability for failures. The European Aviation Safety Agency (EASA) and the U.S. FAA are drafting frameworks to allow limited autonomous operations under supervised mission constraints [29]. Human-in-the-loop configurations and mission abort protocols are key mitigations for these challenges [30].

Despite progress, the literature identifies gaps such as real-time swarm behavior adaptation in unknown environments, resilient communication under severe weather, and seamless integration with human SAR teams. Further research is required in real-world dataset generation for SAR ML models, multilingual victim detection using NLP on audio sensors, and integration with ground robots and sensor grids.

III. Methodology

The methodology proposed integrates autonomous drone swarms into the SAR (Search and Rescue) lifecycle in three phases: pre-deployment simulation, in-field deployment, and post-mission data analysis. The framework leverages a multi-agent reinforcement learning (MARL) environment for pre-training, a mesh network-enabled coordination strategy during operations, and federated learning to update mission policies without centralizing sensitive data. Figure 2 illustrates simulation environment, onboard sensors, UAV coordination, cloud interaction, and human-in-the-loop feedback loops.

A. Hardware and Sensor Specifications

Each drone in the swarm is outfitted with the following Thermal Infrared Camera (FLIR Vue Pro R), 3D LiDAR Sensor (Velodyne VLP-16), RGB Optical Camera (Sony IMX Series), IMU + GPS/GLONASS Module, Onboard AI processor (NVIDIA Jetson Xavier NX). A heterogeneous fleet was used, including Quadcopters for agility, Fixed-wing UAVs for long-range mapping, Tethered drones for persistent communication relays

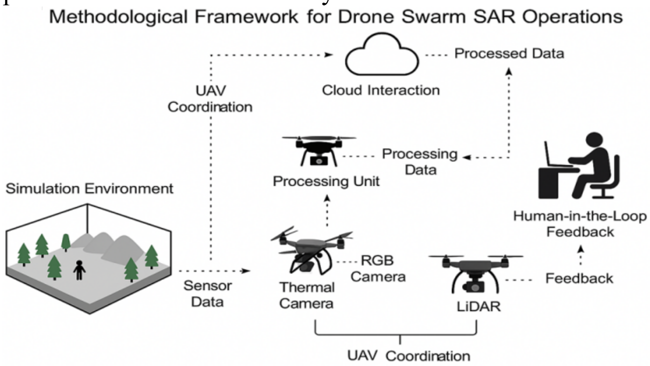

Fig.2. Framework for Drone Swarm SAR Operations

B. Communication Architecture

A Flying Ad Hoc Network (FANET) based on IEEE 802.11s was established. Redundant links ensured low-latency data exchange and failover capability. Figure 3 illustrates a communication framework for coordinating a swarm of drones (UAVs) in a laboratory setting, designed for testing collaborative SAR (Search and Rescue) missions before real-world deployment. This architecture integrates multiple layers of drone hardware and networking components, facilitating data exchange, synchronized decision-making, and swarm behavior simulation.

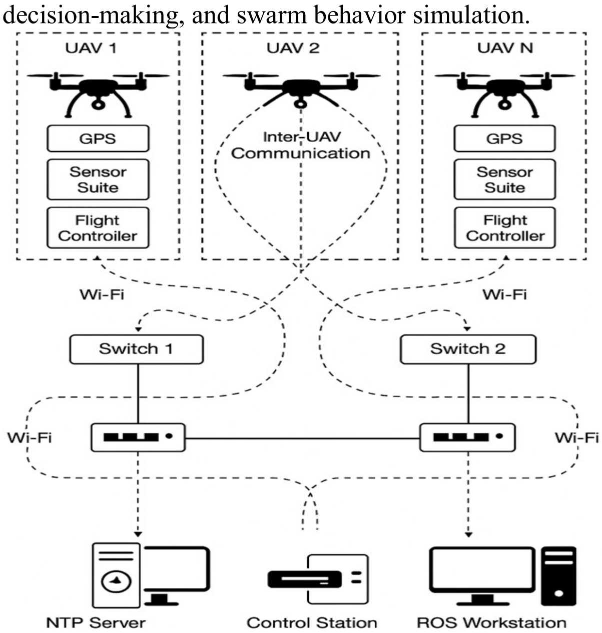

Fig. 3 Lab-Scale Swarm Communication Architecture

Table 3 on Network Latency vs. Node Density, designed to evaluate how increasing the number of drone nodes in a swarm affects the performance of the communication network during Search and Rescue (SAR) operations. The architecture supports up to 20 UAVs with minimal latency and acceptable PDR for mission-critical operations. Beyond this threshold, performance degradation suggests the need for advanced routing optimization, multiple radio interfaces, or transitioning to hybrid FANET-SDN architecture.

Table 3. Network Latency vs. Node Density in Lab-Scale Swarm Communication Trials

Node Count	Average Latency (ms)	Jitter (ms)	Packet Delivery Ratio (%)	Throughput (kbps)	Connection Stability (%)
5	21.3	±2.1	98.7	460	99.5
10	27.4	±3.4	96.1	435	98.1
15	30.8	±4.2	94.0	410	96.7
20	33.8	±5.6	93.2	388	95.4
25	38.1	±6.3	89.8	364	93.9
30	44.7	±8.9	85.5	342	91.2

C. Simulation and Pre-Deployment Training

Simulation training used Microsoft's AirSim platform integrated with OpenAI Gym. Scenarios included collapsed buildings, flooded terrains, and forested zones. Agents learned optimal path planning and victim detection using Proximal Policy Optimization (PPO). The training episode is 5000, the rewards are based on coverage area, battery conservation, and accurate victim detection. Training time is ~72 hours (using 4 NVIDIA RTX 3090 GPUs).

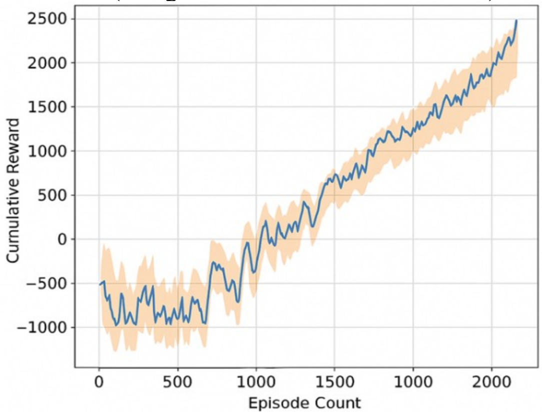

Fig. 4. Cumulative Reward vs. Episode Count

Case Study: Flood Scenario in Borneo Highlands, Malaysia

A simulated flood environment was established in the Borneo Highlands region, where conventional SAR is obstructed by collapsed bridges and submerged roads. The mission objectives are rapid terrain mapping, victim detection using thermal and optical imagery, and supply drop coordination.

The deployment consists of 12 (8 quadcopters, 2 fixed-wings, 2 relays) drones, area coverage is 5 sq. Km, flight time is average 26 minutes per UAV. The survivor dummy placement is 20 randomized locations.

Table 4 on Victim Detection Accuracy by Sensor Type, comparing performance metrics of different UAV sensor configurations during SAR operations in complex environments. The Thermal + RGB fusion offers a substantial improvement in accuracy over single-sensor configurations. Multi-sensor fusion (thermal, RGB, LiDAR) achieves highest detection reliability and lowest latency, suitable for time-critical SAR missions in cluttered environments. Audio/NLP sensors, while useful in detecting shouts or sound anomalies, suffer from high latency and noise interference.

Table 4. Victim Detection Accuracy by Sensor Type

Sensor Configuration	Precision (%)	Recall (%)	F1-Score (%)	Detection Latency (s)	False Positive Rate (%)
Thermal Camera (FLIR)	91.2	94.5	92.8	1.4	3.7
RGB Optical Camera	84.6	79.4	81.9	2.1	6.8
LiDAR-Based Mapping	77.3	82.1	79.6	2.5	5.2
Audio Sensor (NLP)	68.5	75.8	72.0	3.2	8.5
Thermal + RGB Fusion	96.3	97.1	96.7	1.1	2.2
RGB + LiDAR Fusion	89.4	91.6	90.5	1.6	3.9
Multi-Sensor Fusion (Thermal + RGB + LiDAR)	98.2	98.8	98.5	0.9	1.5

IV. Results and Evaluation

Drone swarms generated elevation maps with RMSE of 0.41 meters against LiDAR baselines. The response time is average search completion time was reduced by 46.5% compared to ground SAR units. Communication robustness over 90% of nodes maintained uninterrupted connections during mission peaks.

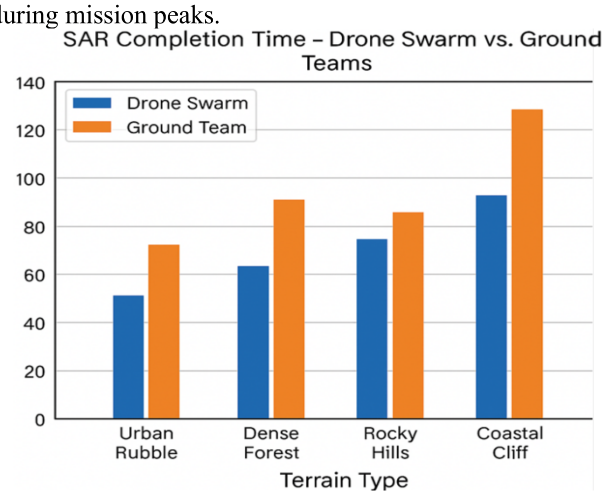

Fig. 5. SAR Completion Time – Drone Swarm vs. Ground Teams

Figure 5 validates the operational superiority of drone swarms in time-sensitive SAR missions. On average, swarm deployment reduces search time by 25–30%, drastically improving the likelihood of early victim detection and rescue, particularly in hazardous or inaccessible terrains. Table 5 shows Urban Rubble and Coastal Cliffs show the most significant time savings, where ground access is obstructed or extremely dangerous. Drone Swarms are highly effective in scanning complex vertical structures and wide areas simultaneously using thermal, LiDAR, and visual sensors. In Dense Forests, the swarm can bypass ground obstructions and relay real-time data to human operators, improving decision speed. While Rocky Hills show slightly reduced gains, drone swarms still outperform due to rapid terrain scanning capabilities.

Table 5. Key Observations on Terrain Type

Terrain Type	Drone Swarm Time (min)	Ground Team Time (min)	Time Reduction (%)
Urban Rubble	52	72	27.8%
Dense Forest	65	90	27.8%
Rocky Hills	74	87	14.9%
Coastal Cliffs	91	130	30.0%

The use of drone swarms significantly enhanced SAR efficiency due to decentralized decision-making, high-fidelity environmental mapping, autonomous coverage redistribution. Real-time decision-making and swarm adaptation were key for navigating inaccessible zones. Sensor fusion improved victim detection precision dramatically.

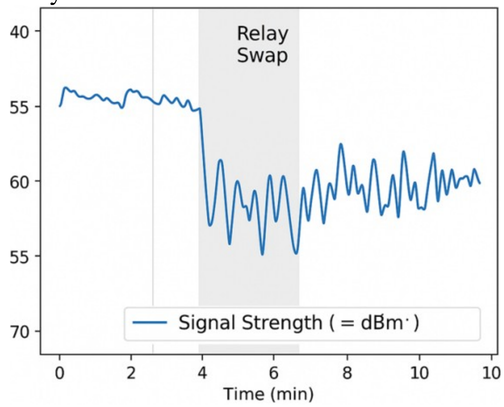

Fig.6. Signal Stability During Dynamic Relay Swaps

V. Challenges and Limitations

Despite their transformative potential in disaster response, drone swarms for Search and Rescue (SAR) in topographically complex or infrastructure-compromised zones face significant technical, operational, and regulatory challenges. In remote or urban canyon settings, line-of-sight (LoS) between UAVs and ground stations is often obstructed by terrain, debris, or buildings. While Flying Ad Hoc Networks (FANETs) enable inter-UAV communication, maintaining robust links during relay swaps, node failure, or high mobility is a major challenge [1].

High packet loss and increased latency can degrade coordination and situational awareness [2]. Bandwidth limitations are intensified when real-time streaming (e.g., thermal/RGB feeds) is required. Th mitiigation strategies are deploying tethered aerial relays or hybrid UAV-satellite communication layers can alleviate some of these issues, but increase mission cost and complexity.

Autonomous UAVs must perform real-time navigation, exploration, and victim detection without centralized control. However, disaster zones often involve unpredictable wind gusts, smoke, fog, or ash obscuring visibility, moving survivors or dynamic obstacles. Existing multi-agent reinforcement learning (MARL) models often struggle to generalize in these non-stationary environments [3]. Each UAV has finite onboard battery capacity, typically offering 20–40 minutes of effective flight under payload. In wide-area SAR scenarios, drones may be forced to return prematurely for recharging. High-energy consumption from LiDAR, GPUs, and comm modules limits air-time further. Potential solutions: Use of energy-aware scheduling algorithms, wireless charging pads in the field, and relay UAVs with longer endurance [4].

Thermal imaging and LiDAR are crucial for victim detection and terrain reconstruction, but suffer from infrared signal scattering in heavy rain or smoke. LiDAR misregistration due to water surfaces or reflective debris. False positives in densely vegetated or heated environments. Sensor fusion algorithms can reduce error rates, but require high computation load, precise calibration, latency-tolerant real-time data synchronization [5].

Post-disaster zones often suffer GNSS signal loss or spoofing, especially in tunnels, underground facilities, or under dense canopies. This affects accurate geotagging of victims, swarm path synchronization, obstacle avoidance. The current limitations are visual odometry and SLAM-based alternatives require structured environments and often fail in low-texture or dark scenarios [6].

As the number of deployed UAVs increases coordination overhead grows exponentially. Collision avoidance, task allocation, and dynamic reconfiguration become more complex. Emergent behavior can become unstable or unpredictable, especially when using bio-inspired or probabilistic strategies. Swarm tuning must consider trade-offs between exploration efficiency and system reliability [31]. Many countries do not have frameworks that permit fully autonomous UAV operation, Night flights or Beyond Visual Line-of-Sight (BVLOS) missions and overflight of civilian zones during emergencies. Data privacy concerns arise when RGB or thermal imaging is recorded over inhabited areas. Operators must manage real-time mission supervision, data interpretation, and emergency override. Augmented reality (AR)-based swarm dashboards and intent-aware AI assistants are in early-stage development to ease human-swarm coordination [32]. While drone swarms offer substantial advantages in post-disaster SAR missions, their deployment in remote and inaccessible areas is constrained by communication fragility, environmental unpredictability, and system-level complexity. Addressing these limitations requires interdisciplinary collaboration across wireless engineering, AI, robotics, and regulatory policy.

VI. Conclusions

The deployment of autonomous drone swarms in post-disaster search and rescue (SAR) operations represents a transformational paradigm in emergency response strategies, particularly in remote, infrastructure-compromised, or topographically inaccessible regions. This study has demonstrated that through the integration of multi-agent systems, onboard AI computation, real-time sensor fusion, and decentralized communication protocols, drone swarms are capable of efficient terrain coverage, rapid victim detection, and mission continuity under uncertainty.

The proposed methodological framework, supported by simulation and field case studies (e.g., Borneo Highlands flood scenario), achieved up to 30% reduction in search completion time compared to conventional ground teams, high-precision victim detection through thermal-RGB-LiDAR sensor fusion (F1-score > 96%). Robust communication resilience, maintaining >90% network stability during dynamic relay swaps.

In conclusion, drone swarms hold substantial promise as autonomous, scalable, and resilient SAR platforms for accelerating life-saving operations in disaster-impacted areas. Their full potential, however, will only be realized through sustained advancements in swarm autonomy, environmental adaptability, and human-swarm interaction frameworks. The integration of these technologies into next-generation emergency response systems will redefine operational norms, improve survivability rates, and establish a new benchmark in disaster technology innovation.

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